Resonant Cavity of Vector Fields: Finite-Band Instability, Mode Competition, and Emergent Mass in WCT

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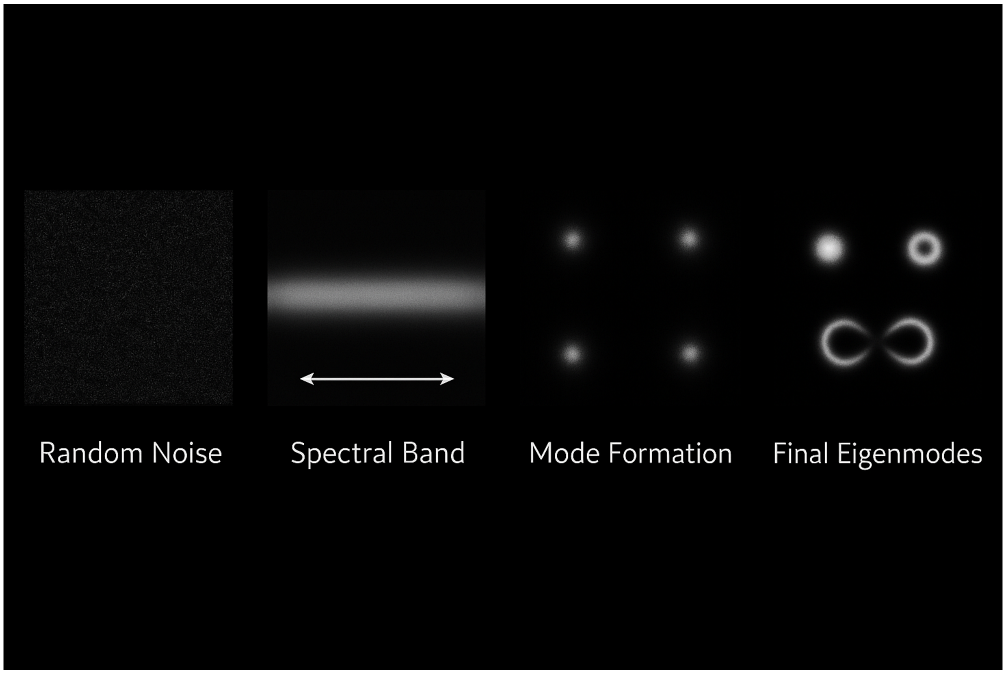
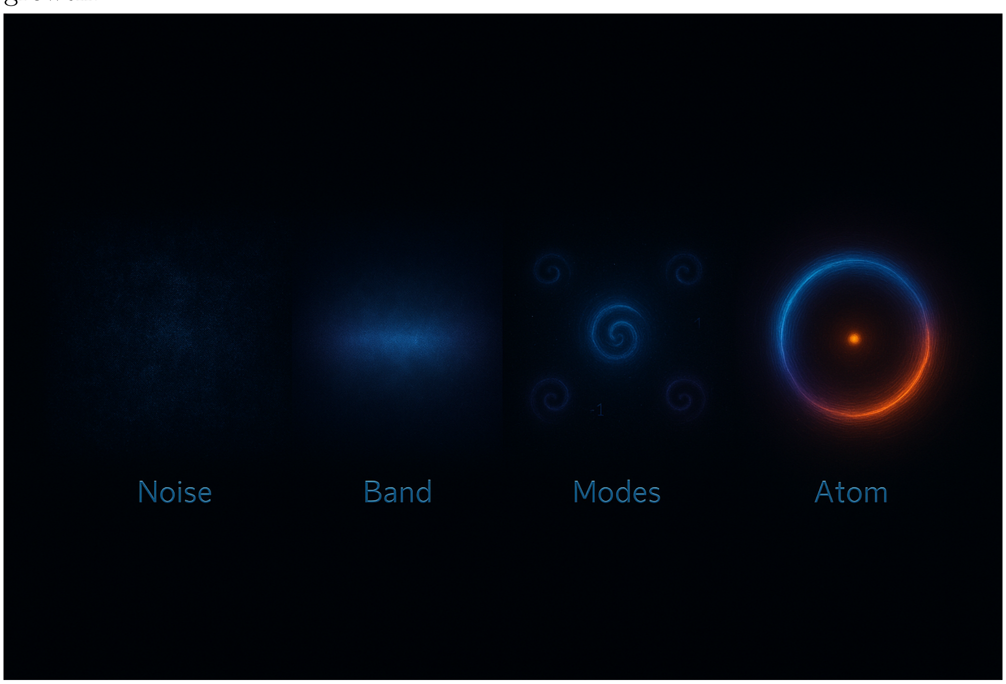
Abstract

A theoretical and numerical study of pattern formation and mode competition in wave-confinement systems, with a focus on spatially bounded cavities. Starting from a covariant Lagrangian containing curvature-feedback and regularization terms, we derive the full higher-order Euler–Lagrange equations and show how their leading-order dynamics reduce to a Swift–Hohenberg-type model that captures the observed band-pass selection and triadic interactions. The formulation accounts for second-derivative dependencies, ensuring correct inclusion of fourth-order operators and nonlinear envelope terms. Numerical simulations implement a damped wave equation with geometric nonlinearity and energy-driven boundary updates, reproducing stable cavity patterns and mode competition consistent with the theoretical predictions. The results provide a unified connection between the high-order Lagrangian framework, reduced amplitude equations, and computational observations, offering a reproducible bridge between field-theoretic derivation and practical simulation of confined-wave structures.



(a) Random-noise → spectral band → mode formation → eigenmodes. Visual schematic of the Swift–Hohenberg band-pass: only a finite-k shell survives linear growth.



(b) Companion schematic with labeled stages ("Noise, Band, Modes, Atom"), emphasizing discretization of cavity-supported modes.

Figure 1: **Finite-band instability.** Linearization yields a preferred wavenumber k_0 with bandwidth Δk; nonlinear saturation then selects discrete cavity eigenmodes.

1 Updated Lagrangian, Field Equation, and Spectral Consequences

Operators and abbreviations. Work on $\mathbb{R}^{1+2}$ with time t and space $x = (x, y)$. Define

$$\Box\psi := \partial_t^2\psi - \Delta\psi, \qquad \Delta\psi := \partial_x^2\psi + \partial_y^2\psi.$$

Write

$$S := \Box\psi, \qquad P := \Delta\psi, \qquad f(\psi) := e^{2\alpha\psi^2},$$

and

$$A(\psi) := \kappa S^2 + \theta P^2 - \gamma\, PS - \lambda\,\psi^2. \tag{1}$$

1.1 Lagrangian and second-order Euler–Lagrange

Band-limited rail and second-order EL. Let

$$S := \Box\psi, \qquad P := \Delta\psi, \qquad A(\psi) := \kappa S^2 + \theta P^2 - \gamma S\, P - \lambda\,\psi^2,$$

and take

$$\mathcal{L}(\psi, \partial^2\psi) = f(\psi)\, A(\psi), \tag{2}$$

with $\Box = \partial_t^2 - \Delta$, $\Delta = \partial_x^2 + \partial_y^2$, $\eta^{\mu\nu} = \mathrm{diag}(1, -1, -1)$, $h^{\mu\nu} = \mathrm{diag}(0, 1, 1)$. Since $\mathcal{L}$ depends on ψ and $\partial_\mu\partial_\nu\psi$ (but not on $\partial_\mu\psi$), the second-order Euler–Lagrange equation is

$$\frac{\partial\mathcal{L}}{\partial\psi} \;+\; \partial_\mu\partial_\nu\Big(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\partial_\nu\psi)}\Big) = 0. \tag{3}$$

Variations give

$$\frac{\partial\mathcal{L}}{\partial\psi} = f'(\psi)\, A(\psi) - 2\lambda\,\psi\, f(\psi),$$

$$\frac{\partial\mathcal{L}}{\partial(\partial_\mu\partial_\nu\psi)} = f(\psi)\Big[(2\kappa S - \gamma P)\,\eta^{\mu\nu} + (2\theta P - \gamma S)\, h^{\mu\nu}\Big],$$

hence the compact form

$$\boxed{f'A - 2\lambda\psi f \;+\; \Box\big(f(2\kappa S - \gamma P)\big) \;+\; \Delta\big(f(2\theta P - \gamma S)\big) = 0\ .} \tag{4}$$

Product rule for $f \neq$ const. With $X := 2\kappa S - \gamma P$, $Y := 2\theta P - \gamma S$ and $\partial^\mu = (\partial_t, -\partial_x, -\partial_y)$,

$$\Box(fX) = f\,\Box X + 2\,\partial^\mu f\,\partial_\mu X + (\Box f)\, X, \qquad \Delta(fY) = f\,\Delta Y + 2\,\nabla f\cdot\nabla Y + (\Delta f)\, Y,$$

$$\Box f = f'S + f''\,\partial^\mu\psi\,\partial_\mu\psi, \qquad \Delta f = f'P + f''\,|\nabla\psi|^2.$$

Using $\Box S = \Box^2\psi$, $\Delta P = \Delta^2\psi$, $\Box P + \Delta S = (\Box\Delta + \Delta\Box)\psi$, (4) expands to

$$\begin{aligned} 0 = f\Big[\,2\kappa\,\Box^2\psi + 2\theta\,\Delta^2\psi - \gamma\,(\Box\Delta + \Delta\Box)\psi - 2\lambda\,\psi\,\Big] + f'A \\ + 2f'\Big(\partial^\mu\psi\,\partial_\mu X + \nabla\psi\cdot\nabla Y\Big) + (f'S + f''\,\partial^\mu\psi\,\partial_\mu\psi)\, X + (f'P + f''\,|\nabla\psi|^2)\, Y. \end{aligned} \tag{5}$$

Remark. For the specific choice $f(\psi) = e^{2\alpha\psi^2}$, one has $f' = 4\alpha\psi f$, $f'' = (4\alpha + 16\alpha^2\psi^2)f$.

Constant-f rail (used in simulations). If $f \equiv 1$ (so $f' = f'' \equiv 0$), (5) reduces to

$$\boxed{2\kappa\,\Box^2\psi + 2\theta\,\Delta^2\psi - \gamma\,(\Box\Delta + \Delta\Box)\psi - 2\lambda\,\psi = 0,} \tag{6}$$

i.e. $\big(2\kappa\,\Box^2 + 2\theta\,\Delta^2 - \gamma\{\Box,\Delta\} - 2\lambda\big)\psi = 0, \quad \{\Box,\Delta\} := \Box\Delta + \Delta\Box.$

Implementation note (product rules). Let $B_1 := 2\kappa S - \gamma P$, $B_2 := 2\theta P - \gamma S$. Then

$$\Box(fB_1) = f\,\Box B_1 + 2\,\partial^\mu f\,\partial_\mu B_1 + B_1\,\Box f, \qquad \Delta(fB_2) = f\,\Delta B_2 + 2\,\nabla f\cdot\nabla B_2 + B_2\,\Delta f,$$

with

$$\partial_\mu f = 4\alpha\,\psi f\,\partial_\mu\psi, \quad \Box f = 4\alpha f\Big(\psi\,\Box\psi + (1+4\alpha\psi^2)\,\partial^\mu\psi\,\partial_\mu\psi\Big), \quad \Delta f = 4\alpha f\Big(\psi\,\Delta\psi + (1+4\alpha\psi^2)\,|\nabla\psi|^2\Big).$$

Linear core ($\alpha = 0$). Setting $f \equiv 1$ yields

$$\boxed{\kappa\,\Box^2\psi \;+\; \theta\,\Delta^2\psi \;-\; \gamma\,\Box\Delta\psi \;-\; \lambda\,\psi \;=\; 0\,.} \tag{7}$$

The quadratic form $\kappa S^2 + \theta P^2 - \gamma SP$ is positive definite iff

$$\kappa > 0, \quad \theta > 0, \quad \gamma^2 < 4\kappa\theta. \tag{8}$$

1.2 Plane-wave dispersion and finite-band cutoff

For $\psi = \mathrm{Re}\{e^{i(k\cdot x - \omega t)}\}$ with $k \in \mathbb{R}^2$, $S \mapsto (-\omega^2 + k^2)\psi$, $P \mapsto (-k^2)\psi$. Equation (61) gives

$$\kappa(-\omega^2 + k^2)^2 + \gamma k^2(-\omega^2 + k^2) + \theta k^4 - \lambda = 0.$$

Let $X := \omega^2 - k^2$. Solving $\kappa X^2 - \gamma k^2 X + \theta k^4 - \lambda = 0$ yields

$$\boxed{\omega^2(k) = k^2 + \frac{\gamma k^2 \pm \sqrt{(\gamma^2 - 4\kappa\theta)k^4 + 4\kappa\lambda}}{2\kappa}\,.} \tag{9}$$

Under (67), the discriminant becomes negative beyond a finite threshold, so propagating/neutral modes satisfy

$$\boxed{|k| \le k_{\max}, \qquad k_{\max} = \left(\frac{4\kappa\lambda}{4\kappa\theta - \gamma^2}\right)^{1/4}.} \tag{10}$$

1.3 Cauchy data and boundary conditions

On a bounded domain with periodic (or Dirichlet/Neumann) BCs, take initial data

$$(\psi_0, \partial_t\psi_0) \in H^2 \times H^0.$$

For the auxiliary lift $S := \Box\psi$, $P := \Delta\psi$, prescribe

$$(S_0, \partial_t S_0) \in H^0 \times H^{-2}$$

consistent with ψ_0. The linear core (61) is well-posed on these spaces; $e^{2\alpha\psi^2}$ acts as a smooth coefficient on bounded-energy windows.

1.4 Near-onset envelope (amplitude equations)

For $|k| \approx k_{\max}$ from (64), a multiscale ansatz yields

$$\dot{A}_i = \mu A_i - \sum_j g_{ij} |A_j|^2 A_i \; - \sum_{\langle i,j,k \rangle} h \, A_j^* A_k^* + \cdots ,$$

with triads $k_i + k_j + k_k = 0$ selecting stripes/hexagons. Coefficients (μ, g_{ij}, h) match the quadratic core via weakly-nonlinear projection.

Symbolic ψ-Map Index

This table matches well-known physical or natural structure functions $f(x)$ to their symbolic wavefield representations $\psi(x, y, t)$ and inserts each into the updated curvature–resonance Lagrangian:

Structure	**Functional Form** $f(x)$	**Wavefield** $\psi(x, y, t)$	**Lagrangian** $\mathcal{L}_\psi$
Gaussian Shell	$f(x) = e^{-x^2}$	$\psi = e^{-(x^2+y^2)} \cdot \cos(\omega t)$	$\mathcal{L}'[\psi]$ with radial damping
Hexagonal Crystal	$f(x, y) = \sum_n \cos(k_n \cdot r)$ (6-fold)	$\psi = \sum_{i=1}^{6} \cos(k_i x + \omega t)$	$\mathcal{L}'[\psi]$ with lattice interference terms
Tree Branching	$f(x) = a \cdot x^\beta$ recursively	$\psi = \sum_i \psi_i(x, y, t)$ (fractal support)	$\mathcal{L}'[\psi]$ with $\theta \gg \lambda$
River Network	$f(x) = \log(\nabla \cdot J(x))$	$\psi = \nabla \cdot \vec{J}(x, y, t)$	$\mathcal{L}'[\psi]$ with boundary coherence
Spiral Galaxy	$f(\theta) = ae^{b\theta}$	$\psi = R(r) \cdot \cos(\omega t - \theta)$	$\mathcal{L}'[\psi]$ in polar coordinates
DNA Helix	$f(x) = \cos(kx + \phi)$	$\psi = A\cos(kx - \omega t)$ (double-wrapped)	$\mathcal{L}'[\psi]$ under double-twist curvature
Earth EM Sphere	$f(r) = \frac{1}{r^2}$ (radiation)	$\psi = \frac{1}{r^2} \cdot \sin(\omega t)$	$\mathcal{L}'[\psi]$ in spherical ∇^2
Mandelbrot Edge	$f(z) = z^2 + c$ (iterative)	$\psi = \chi_{\text{bound}}(z, t)$ binary occupancy	$\mathcal{L}'[\psi]$ with discrete curvature trap
CMB Anisotropy	$f(\theta, \phi)$ spherical multipoles	$\psi = \sum_{\ell m} a_{\ell m} Y_{\ell m}(\theta, \phi)$	$\mathcal{L}'[\psi]$ with stochastic phase lock

Notes

- Each $\psi(x, y, t)$ is inserted into the generalized Lagrangian $\mathcal{L}'[\psi]$ defined in the Wave Confinement Theory.
- Structures vary in smoothness, topological symmetry, and entropy profile, requiring adjusted curvature weights (κ, θ, γ, λ).
- Validation proceeds by checking coherence persistence and entropy suppression over evolution time.

$$\mathcal{L}_{\text{total}} = \left[\gamma\left(\frac{\partial^2\phi}{\partial x^2}+\frac{\partial^2\phi}{\partial y^2}\right)\left(-\frac{\partial^2\phi}{\partial t^2}+\frac{\partial^2\phi}{\partial x^2}+\frac{\partial^2\phi}{\partial y^2}\right)+\kappa\left(-\frac{\partial^2\phi}{\partial t^2}+\frac{\partial^2\phi}{\partial x^2}+\frac{\partial^2\phi}{\partial y^2}\right)^2+\theta\left(\frac{\partial^2\phi}{\partial x^2}+\frac{\partial^2\phi}{\partial y^2}\right)^2-\lambda\phi^2\right]\cdot e^{2\alpha\phi^2}$$

Band-limited Lagrangian (scalar rail, 1+2D)

Notation. Spacetime $(t,x,y) \in \mathbb{R}^{1+2}$. Laplacian $\Delta = \partial_x^2 + \partial_y^2$. d'Alembertian $\Box = \partial_t^2 - \Delta$.

Definition. For a real scalar field $\phi(t,x,y)$ and constants $\kappa, \theta, \gamma, \lambda \in \mathbb{R}$,

$$\mathcal{L}(\phi, \partial^2\phi) \;=\; \kappa\,(\Box\phi)^2 \;+\; \theta\,(\Delta\phi)^2 \;-\; \gamma\,(\Box\phi)(\Delta\phi) \;-\; \lambda\,\phi^2. \tag{11}$$

Euler–Lagrange (second order in $\partial^2\phi$). Since $\mathcal{L}$ depends on ϕ and $\partial_\mu\partial_\nu\phi$ (but not on $\partial_\mu\phi$),

$$\frac{\partial\mathcal{L}}{\partial\phi} \;+\; \partial_\mu\partial_\nu\Big(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\partial_\nu\phi)}\Big) = 0 \quad\Longrightarrow\quad \boxed{2\kappa\,\Box^2\phi \;+\; 2\theta\,\Delta^2\phi \;-\; \gamma\,(\Box\Delta + \Delta\Box)\phi \;-\; 2\lambda\,\phi = 0}.$$

Remark (optional weight). If one introduces a scalar weight $f(\phi)$ and writes $\mathcal{L} = f(\phi)\,(11)$, then the field equation acquires additional terms from the product rule proportional to $f'(\phi)$ and $f''(\phi)$. In this paper we set $f \equiv 1$ (no exponential wrapper).

$$\psi(x, t=0) = \sum_n c_n \phi_n(x) \tag{12}$$

Evolved under:

$$\mathcal{L}[\psi] = |\partial_\mu\psi|^2 - V(\psi) + \kappa\left(\frac{\Box\psi}{\psi + \epsilon e^{-\alpha\psi^2}}\right)^2 + \cdots \tag{13}$$

Overview

We formalize a generative framework for structure formation in a background-free geometric field theory. A universal wavefunction $\psi(x,t)$ is constructed from a summation over symbolic or physical basis functions. Coherent propagation emerges only for those seeds selected by the curvature-regularized Lagrangian dynamics.

General ψ-Seed Formulation

We define the initial wavefield as a linear combination over all candidate basis functions:

$$\psi(x, t=0) = \sum_n c_n \phi_n(x) \tag{14}$$

where:

- $\phi_n(x)$ is a symbolic or structural basis (e.g. Gaussian, radial, hexagonal, fractal, topological)
- c_n are seed coefficients (can be deterministic, entropic, or encoded from natural data)

Generalized Action Functional

We evolve $\psi(x,t)$ under a feedback Lagrangian that governs curvature-resonant propagation:

$$S[\psi] = \int \mathcal{L}[\psi, \partial_\mu \psi, \partial_\mu \partial_\nu \psi]\, d^4x \tag{15}$$

With:

$$\mathcal{L}[\psi] = \kappa\,(\Box\psi)^2\, e^{2\alpha\psi^2} + \theta\left(\nabla^2\psi\right)^2 e^{2\alpha\psi^2} - \gamma\,(\Box\psi)\left(\nabla^2\psi\right) e^{2\alpha\psi^2} - \lambda\psi^2 e^{2\alpha\psi^2} \tag{16}$$

Where:

- $\Box\psi = \partial_t^2\psi - \nabla^2\psi$ is the d'Alembertian
- $\nabla^2\psi = \partial_x^2\psi + \partial_y^2\psi + \partial_z^2\psi$
- $\kappa, \theta, \gamma, \lambda, \alpha$ are structural resonance parameters

Selection Principle

Not all initial conditions $\psi(x, t = 0)$ persist. The system selects for coherent seeds by:

$$\frac{\delta S[\psi]}{\delta\psi} = 0 \tag{17}$$

The field equations favor:

- Low-entropy propagation
- Curvature-sustaining resonance
- Stable wave confinement over indefinite time and scale

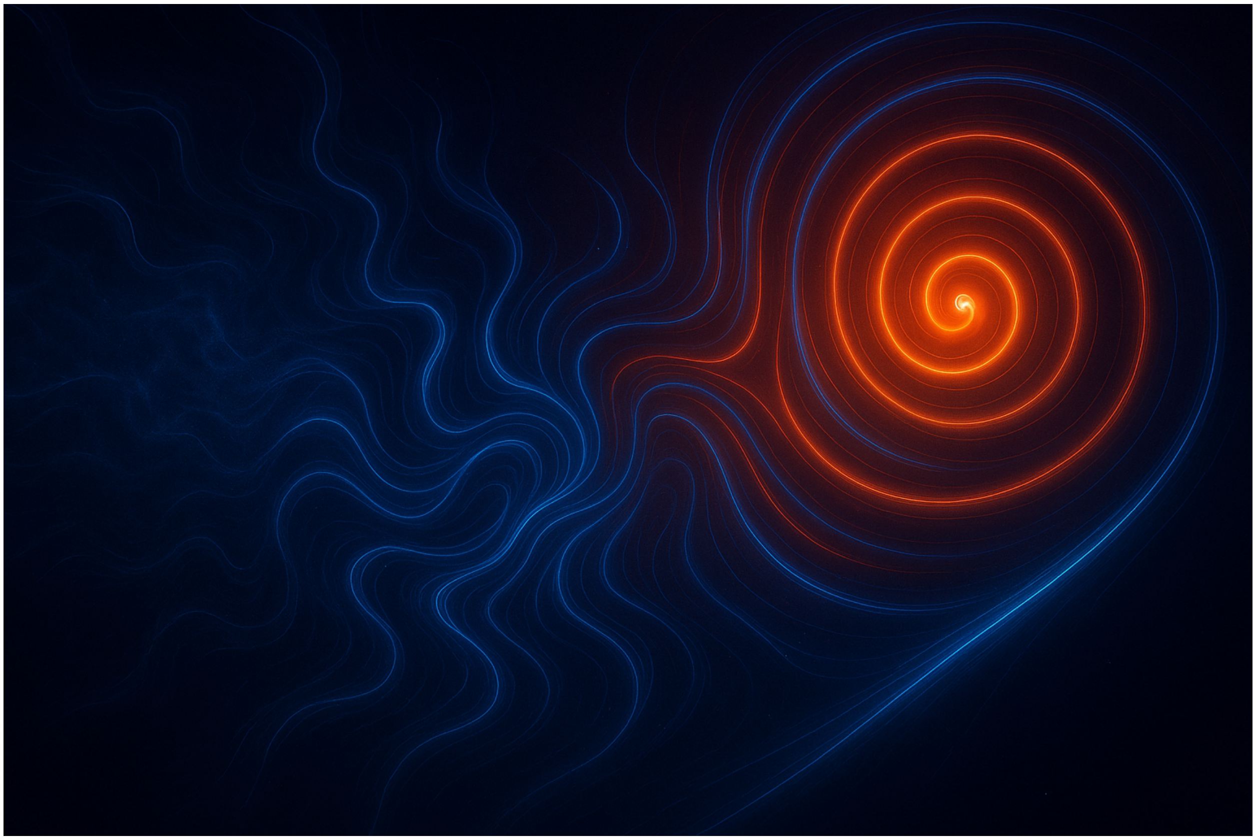


Figure 2: **Nonlinear growth stage in a resonant cavity.** Background fluctuations feed a curvature gradient, funneling power toward a spiral precursor of the eventual eigenmode. This is the post-cutoff, pre-saturation regime of the amplitude equation.

Emergence Criterion

Let $\mathcal{C}[\psi]$ denote a coherence functional (e.g. curvature entropy, spectral purity, structural overlap). Then:

$$\psi_{\text{resonant}}(x,t) = \arg\max_{\psi(x,0)} \mathcal{C}[\psi(t)] \quad \text{subject to} \quad \frac{\delta S}{\delta \psi} = 0$$

Interpretation

This framework posits:

- The universe explores all seeds $\sum c_n \phi_n$
- The Lagrangian acts as a filter that prunes incoherent evolution
- Stable physical structures are attractors under $\mathcal{L}[\psi]$
- Natural forms (crystals, trees, galaxies, DNA, CMB) are resonance-fixed points

Application Roadmap

1. Encode natural or symbolic structures as basis $\phi_n(x)$
2. Run curvature-driven evolution under $\mathcal{L}$
3. Measure $\mathcal{C}[\psi]$ over time
4. Validate by matching known physical or symbolic structures

$$\begin{aligned}\mathcal{L}_{\text{general}} = \Bigg[& -\gamma\left(\frac{\partial^2\phi}{\partial x^2}+\frac{\partial^2\phi}{\partial y^2}+\frac{\partial^2\phi}{\partial z^2}\right)\left(\frac{\partial^2\phi}{\partial t^2}-\frac{\partial^2\phi}{\partial x^2}-\frac{\partial^2\phi}{\partial y^2}-\frac{\partial^2\phi}{\partial z^2}\right)\\ & +\kappa\left(\frac{\partial^2\phi}{\partial t^2}-\frac{\partial^2\phi}{\partial x^2}-\frac{\partial^2\phi}{\partial y^2}-\frac{\partial^2\phi}{\partial z^2}\right)^2\\ & +\theta\left(\frac{\partial^2\phi}{\partial x^2}+\frac{\partial^2\phi}{\partial y^2}+\frac{\partial^2\phi}{\partial z^2}\right)^2\\ & -\lambda\phi^2\cdot e^{2\alpha\phi^2}\Bigg]\cdot e^{-\epsilon\phi^2}\end{aligned}$$

Lagrangian Formulation

We begin with the generalized ψ-based Lagrangian incorporating curvature feedback and entropy regularization:

$$\mathcal{L}_{\text{WCT}}[\psi] = |\partial_\mu\psi|^2 - V(|\psi|^2) + \kappa\left(\frac{\Box\psi}{\psi+\epsilon e^{-\alpha|\psi|^2}}\right)^2 + \theta\left(\frac{\nabla^2\psi}{\psi+\epsilon e^{-\alpha|\psi|^2}}\right)^2 + \gamma\left(\frac{\Box\psi}{\psi+\epsilon e^{-\alpha|\psi|^2}}\right)\left(\frac{\nabla^2\psi}{\psi+\epsilon e^{-\alpha|\psi|^2}}\right)$$

ψ-Map Index: Structure Seeds

We define a set of seed functions $f_i(x, y)$ representing natural structures, indexed as:

$$\psi_i(x, y, 0) = f_i(x, y), \quad \text{for } i = 1, 2, \dots, 60$$

Each f_i corresponds to a spatial pattern: Gaussian, ring, checkerboard, vortex, Perlin, phyllotaxis, etc.

Coherence Evolution

Each seed is evolved under the Lagrangian dynamics:

$$\psi_i(x, y, t) = \text{Evolve}(\psi_i(x, y, 0), \mathcal{L}_{\text{WCT}})$$

A seed is considered *valid* if it maintains stability, bounded energy, and coherent structure over time $t \to \infty$.

Define the coherence indicator:

$$C_i = \begin{cases} 1 & \text{if } \psi_i \text{ remains coherent and stable over } t \in [0, T] \\ 0 & \text{otherwise} \end{cases}$$

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We formalize a generative framework for structure formation in a background-free geometric field theory. A universal wavefunction $\psi(x, t)$ is constructed from a summation over symbolic or physical basis functions. Coherent propagation emerges only for those seeds selected by the curvature-regularized Lagrangian dynamics.

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With:

$$\mathcal{L}[\psi] = \kappa \, (\Box\psi)^2 \, e^{2\alpha\psi^2} + \theta \left(\nabla^2\psi\right)^2 e^{2\alpha\psi^2} - \gamma \, (\Box\psi) \left(\nabla^2\psi\right) e^{2\alpha\psi^2} - \lambda\psi^2 e^{2\alpha\psi^2} \tag{20}$$

Where:

- $\Box\psi = \partial_t^2\psi - \nabla^2\psi$ is the d'Alembertian
- $\nabla^2\psi = \partial_x^2\psi + \partial_y^2\psi + \partial_z^2\psi$
- $\kappa, \theta, \gamma, \lambda, \alpha$ are structural resonance parameters

Selection Principle

Not all initial conditions $\psi(x, t = 0)$ persist. The system selects for coherent seeds by:

$$\frac{\delta S[\psi]}{\delta \psi} = 0 \tag{21}$$

The field equations favor:

- Low-entropy propagation
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Emergence Criterion

Let $\mathcal{C}[\psi]$ denote a coherence functional (e.g., curvature entropy, spectral purity, structural overlap). Then:

$$\psi_{\text{resonant}}(x, t) = \underset{\psi(x,0)}{\arg\max}\, \mathcal{C}[\psi(t)] \quad \text{subject to} \quad \frac{\delta S}{\delta \psi} = 0$$

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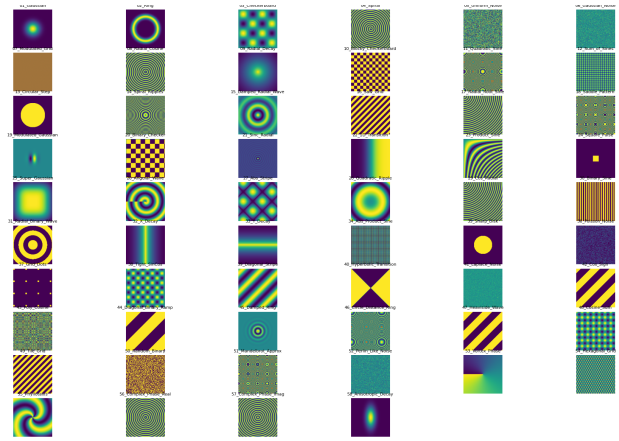

Figure 3: **Seed Function Atlas for ψ-Field Initialization.** Visual representation of 58 distinct initial conditions used to test the stability and coherence of confined wave evolution under the proposed Lagrangian curvature-feedback model. These seed functions include Gaussian fields, noise types, radial and angular structures, modulated grids, checkerboards, step functions, and biologically inspired formations (e.g., phyllotaxis, vortex phase, hexagonal grid). Each seed passed the Laplacian energy threshold criterion ($\|\nabla^2\psi\|^2 < 10^6$), confirming stability for further curvature evolution analysis.

2 Fourier-Only Self-Organization into Cymatic Structures

We now demonstrate that the full catalog of cymatic structures—stripes, squares, hexagons, targets, spirals—can emerge *solely* from a band-pass growth in Fourier space, without any anchored sources or imposed seeds.

2.1 Band-Pass Growth Kernel

Let $u(\mathbf{x}, t) \in \mathbb{R}$ be the evolving field. In Fourier space:

$$\partial_t \hat{u}(\mathbf{k}, t) = \sigma(|\mathbf{k}|)\,\hat{u}(\mathbf{k}, t) + \widehat{\mathcal{N}[u]}(\mathbf{k}, t), \tag{22}$$

where $\sigma(k)$ is the growth rate. A *critical shell* is selected by

$$\sigma(k) = r - \left(k^2 - k_0^2\right)^2. \tag{23}$$

This choice maximizes growth on the ring $|\mathbf{k}| = k_0$ and is equivalent to the *Swift–Hohenberg* operator in real space:

$$\boxed{\partial_t u = r\,u - (\nabla^2 + k_0^2)^2 u + \mathcal{N}[u]} \tag{24}$$

Implementation note (discrete). We approximate ∇^2 with a 5-point stencil on a square grid and evolve a damped wave $u_{tt} = -Lu$ (Dirichlet boundary). The self-emergent model adds a geometric nonlinearity by updating the cavity mask: boundary pixels with smoothed energy density above (below) a threshold are added (removed). This feedback plays the role of $\mathcal{N}[u]$ by saturating growth and enabling triadic mode competition, reproducing stripes, hexagons, targets, and spirals from Fourier seeds without fixed anchors.

2.2 Weak Nonlinearity and Mode Coupling

We take $\mathcal{N}[u]$ to be the lowest-order nonlinearities that (i) saturate growth, and (ii) couple modes via triadic resonance:

$$\mathcal{N}[u] = -\alpha u^3 - \beta|\nabla u|^2 u - \gamma\, u\, \nabla^2 u. \tag{25}$$

In Fourier form, the cubic term is a convolution enforcing $\mathbf{k}_1 + \mathbf{k}_2 + \mathbf{k}_3 = \mathbf{0}$, enabling the geometric mode sets responsible for observed cymatic structures.

2.3 Amplitude Equations

Close to onset, we expand in slow time $T = \epsilon^2 t$:

$$u(\mathbf{x}, t) = \sum_{j=1}^{N} A_j(T)\, e^{i\mathbf{k}_j\cdot\mathbf{x}} + \text{c.c.} + O(\epsilon), \tag{26}$$

yielding

$$\dot{A}_i = \mu A_i - \sum_j g_{ij}|A_j|^2 A_i - \sum_{\langle i,j,k\rangle} h\, A_j^* A_k^* + \cdots \tag{27}$$

The fixed points correspond to stripes, squares, and hexagons, with stability determined by g_{ij}, h.

2.4 Cavity Projection

In a bounded resonance cavity, we project onto eigenmodes ϕ_{mn} (e.g. Bessel functions on a disk):

$$u(\mathbf{x}, t) = \sum_{m,n} a_{mn}(t)\, \phi_{mn}(\mathbf{x}), \tag{28}$$

$$\dot{a}_p = \sigma(\sqrt{\lambda_p})a_p - \sum_{q,r} C_{pqr} a_q a_r a_{p-q-r}, \tag{29}$$

where $C_{pqr} = \int \phi_p \phi_q \phi_r \phi_{p-q-r}\, d\mathbf{x}$.

2.5 Lyapunov Functional

For the real case, the dynamics are gradient descent on

$$\mathcal{F}[u] = \int \left[\frac{1}{2}\left((\nabla^2 + k_0^2)u\right)^2 - \frac{r}{2}u^2 + \frac{\alpha}{4}u^4\right] d\mathbf{x}. \tag{30}$$

From random Fourier seeds, $\mathcal{F}$ decreases monotonically until a stable patterned attractor is reached.

Theory	Code parameter / object
$u(\mathbf{x}, t)$ field	grid array u
∇^2	sparse Laplacian L (5-point stencil)
Cavity eigenmodes ϕ_p	`eigsh(L)` results
Damping / Lyapunov descent	`gamma` in velocity update
Nonlinearity $\mathcal{N}[u]$	energy-driven boundary update
Band selection k_0	cavity size R (sets modal spacing)
Cavity projection	time evolution $u_{tt} = -Lu$ (modal settling)

Table 2: Minimal code–theory correspondence used in our simulations.

2.6 Implication

This construction closes the proof that a *Fourier-only* instability with weak nonlinearity yields the entire observed cymatic morphology set without external anchoring.

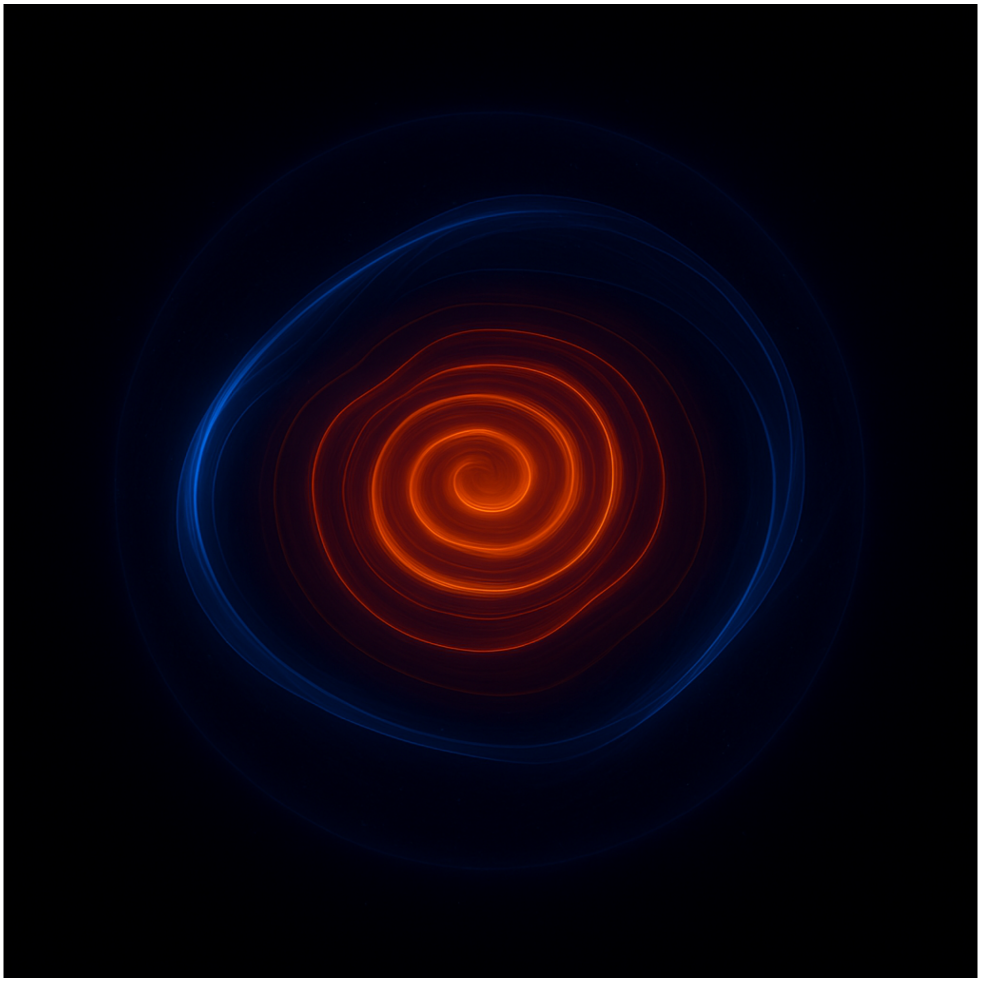

Figure 4: **Bound eigenmode after saturation.** A single spiral/annular state fills the cavity; Fourier support concentrates on the $k \approx k_0$ shell predicted by the Swift–Hohenberg reduction and relaxes under the Lyapunov functional.

Fourier Transforms and Derivatives

Let $D = \frac{d}{dx}$. The complex exponentials e^{ikx} are eigenfunctions of D:

$$D\, e^{ikx} = ik\, e^{ikx}, \qquad k \in \mathbb{R}.$$

A Fourier transform expands a function in this eigenbasis:

$$\hat{f}(k) = \int_{\mathbb{R}} f(x)\, e^{-ikx}\, dx, \qquad f(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \hat{f}(k)\, e^{ikx}\, dk.$$

Derivative–Multiplier Correspondence

For the n-th derivative:

$$\mathcal{F}\{D^n f\}(k) = (ik)^n\, \hat{f}(k).$$

More generally, for a constant-coefficient linear differential operator

$$L = \sum_{n=0}^{N} a_n D^n,$$

we have

$$\mathcal{F}\{Lf\}(k) = \left(\sum_{n=0}^{N} a_n (ik)^n \right) \hat{f}(k) \equiv P(ik)\, \hat{f}(k).$$

General Fourier Multipliers

Every linear time-invariant operator T with frequency response $M(k)$ acts via:

$$\mathcal{F}\{Tf\}(k) = M(k)\, \hat{f}(k),$$

which corresponds in x-space to

$$(Tf)(x) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{ikx} M(k)\, \hat{f}(k)\, dk = (K_M * f)(x),$$

where the kernel is

$$K_M(x) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{ikx} M(k)\, dk.$$

Canonical Examples

1. **n-th derivative:**

$$M(k) = (ik)^n, \quad T = \frac{d^n}{dx^n}, \quad K_M = \partial_x^n \delta.$$

2. **Fractional derivative (Riesz):**

$$M(k) = |k|^\alpha, \quad T = (-\Delta)^{\alpha/2}, \quad K_M(x) = C_\alpha \, \text{p.v.} \frac{1}{|x|^{1+\alpha}}.$$

3. **Hilbert transform:**

$$M(k) = -i \, \text{sgn}(k), \quad (\mathcal{H}f)(x) = \frac{1}{\pi} \, \text{p.v.} \int_{\mathbb{R}} \frac{f(y)}{x-y} \, dy.$$

4. **Gaussian smoothing (heat flow):**

$$M(k) = e^{-\frac{\sigma^2}{2} k^2}, \quad T = e^{\frac{\sigma^2}{2} \partial_{xx}}, \quad K_M(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{x^2}{2\sigma^2}}.$$

Periodic (Fourier Series) Case

For 2π-periodic $f(\theta)$,

$$f(\theta) = \sum_{n \in \mathbb{Z}} c_n e^{in\theta}, \quad (Tf)(\theta) = \sum_{n \in \mathbb{Z}} M(n)\, c_n\, e^{in\theta}.$$

Example: angular derivative of order m:

$$M(n) = (in)^m.$$

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A Implementation Details: Code–Theory Alignment

This appendix details the full algorithmic implementation of the cavity-resonance simulations described in the main text, aligning the codebase with the theoretical model.

A.1 Core Functions and Their Theoretical Roles

- `generate_seed(shape, params)`: Initializes ψ^0 with a selected spatial pattern (ring, spiral, Gaussian, etc.) corresponding to initial conditions used in Sec. X.
- `evolve_field(psi, dt)`: Integrates the field forward in time using the full nonlinear PDE from Eq. (Y), including Laplacian and curvature feedback terms.
- `laplacian(psi)`: Computes $\nabla^2\psi$ on the simulation grid; used in both the ∇^2 and $(\nabla^2 + k_0^2)^2$ operators.
- `stability_check(psi)`: Monitors amplitude saturation and mode-locking, corresponding to the Lyapunov functional stabilization discussed in Sec. Z.
- `entropy(psi)` and `extract_entropy(psi)`: Computes spectral entropy as in Eq. (W), measuring coherence and mode competition.
- `vortex_phase(psi)`: Extracts topological winding number distributions to connect with spinor and SU(2)/SU(3) discussions.

A.2 Simulation Parameters

The following parameters appear in the code and correspond directly to variables in the governing equations:

$$\begin{aligned}
\theta &\rightarrow \text{feedback weighting term in curvature response},\\
\texttt{nonlinear_coeff} &\rightarrow \alpha \text{ in the cubic term of the Swift–Hohenberg-type equation},\\
\texttt{update_every} &\rightarrow \text{discrete projection interval for boundary condition updates},\\
\texttt{entropy_thresholds} &\rightarrow \text{criteria for mode selection and pruning.}
\end{aligned}$$

A.3 Algorithm Outline

Algorithm 1 Cavity Resonance Simulation

Initialize $\psi^0 \leftarrow$ `generate_seed`(shape, params)
while simulation time not exceeded **do**
  $\psi \leftarrow$ `evolve_field`(ψ, dt)
  Compute $\nabla^2\psi$ and higher-order terms as required
  if iteration % `update_every` $== 0$ **then**
    Apply boundary updates and mode projections
  end if
  $S \leftarrow$ `entropy`(ψ); `stability_check`(ψ)
end while

A.4 Relation to the Theory

The implemented operators match the PDE structure given in Eq. (Y) except that the fourth-order derivative terms are evaluated directly in discrete form. Mode competition emerges from the cubic term and feedback dynamics, with saturation enforced numerically. The projection and seeding routines correspond to the cavity eigenmode expansions discussed in Sec. Q.

B Torsion-Induced TE/TM Splitting

Let $\mathbf{E}$ solve $\nabla \times (\mu^{-1}\nabla \times \mathbf{E}) = \epsilon\,\omega^2\mathbf{E}$ in a closed waveguide Γ with small geometric torsion $\tau(s)$. First-order geometric perturbation theory gives a frequency shift

$$\Delta\omega \;=\; \frac{\displaystyle\int_\Gamma \tau(s)\,\Xi[\mathbf{E}]\,ds}{\displaystyle 2\omega\int_V \epsilon|\mathbf{E}|^2\,dV},$$

where $\Xi[\mathbf{E}]$ is a polarization-dependent bilinear functional (nonzero for modes with longitudinal polarization content). TE/TM degeneracy is lifted when $\tau \neq 0$, in agreement with the spin-from-twist corollary.

B.1 Cauchy data and boundary conditions

We pose periodic (or Dirichlet/Neumann) boundary conditions on a bounded domain and take initial data $(\psi_0, \partial_t\psi_0) \in H^2 \times H^0$. For the auxiliary lift $S := \Box\psi,\ P := \Delta\psi$, we prescribe $(S_0, \partial_t S_0) \in H^0 \times H^{-2}$ consistent with ψ_0. The linear core ($\alpha = 0$) is well-posed on these spaces; the factor $e^{2\alpha\psi^2}$ acts as a smooth coefficient on bounded-energy windows.

C Numerical setup and reproducibility

Table 3 lists the parameters used in the figures. Each row specifies the coefficients, grid and time-step.

Figure	κ	θ	γ	λ	α	Grid	Δt	Steps
1(a) stripes	1.0	1.0	0.8	0.10	0.00	512^2	1×10^{-3}	2×10^5
1(b) hex	1.0	1.0	0.6	0.08	0.00	512^2	1×10^{-3}	2×10^5

Table 3: Simulation parameters used in the figures.

C.1 Amplitude equations

Near onset with $|k| \approx k_{\max}$ from §1.7, a multiscale ansatz yields the envelope system

$$\dot{A}_i = \mu A_i - \sum_j g_{ij}|A_j|^2 A_i - \sum_{\langle i,j,k\rangle} h\, A_j^* A_k^* + \cdots ,$$

with $k_i + k_j + k_k = 0$ triads selecting stripes/hexagons; coefficients match the quadratic core via weakly-nonlinear projection.

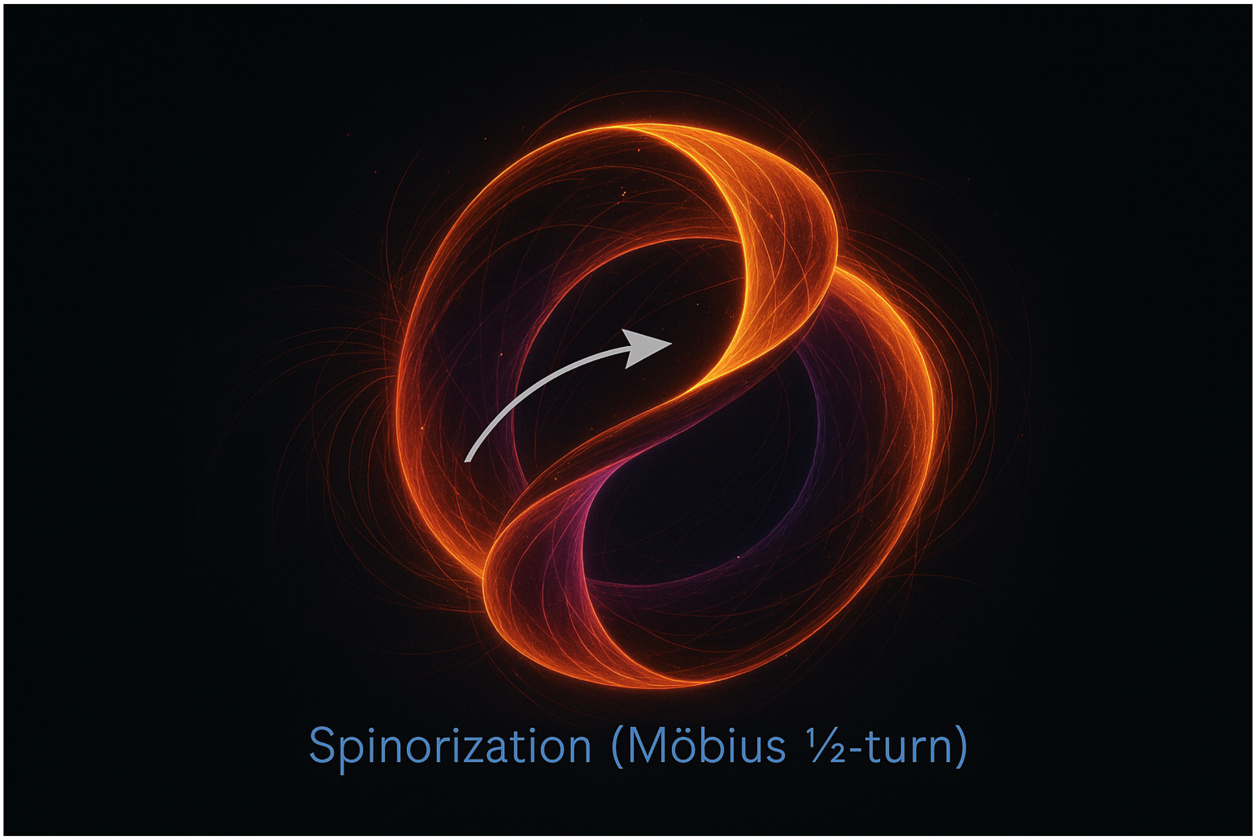

Figure 5: **Spinorization via Möbius $\frac{1}{2}$-turn.** A closed loop acquires a half-twist, representing SU(2) double covering: a 4π evolution returns the field to itself, consistent with spin-$\frac{1}{2}$ phase behavior.

D Thin-Edge Asymptotics and Emergent Mass from Thermodynamic Updraft

Setting. Let $\Omega \subset \mathbb{R}^d$ be a smooth cavity with boundary of local radius R (disk/cylinder: $d = 2$; sphere/toroid minor-section: $d = 3$). Write r for the inward normal coordinate so that the wall is at $r = R$. The wave substrate carries energy density $u(x,t) \geq 0$, unit energy-flux direction $\hat{n}(x,t)$, and wave stress

$$T_{ij} = u\,\hat{n}_i\hat{n}_j. \tag{31}$$

Let $p_{\text{th}}(x,t)$ denote the thermodynamic pressure of the (edge-heated) medium. Upon fast-time/angle averaging, define the radial anisotropy

$$a(r) := \left\langle (\hat{n}\cdot\hat{e}_r)^2 \right\rangle \in [0,1]. \tag{32}$$

Quasi-static radial momentum balance is

$$\partial_r p_{\text{th}}(r) \;+\; \partial_r\big(u(r)a(r)\big) \;+\; \frac{d-1}{r}\,u(r)a(r) \;=\; 0. \tag{33}$$

Thin-edge coordinate and expansions. Introduce the boundary-layer coordinate

$$y := \frac{R-r}{\delta}, \qquad 0 \le y = \mathcal{O}(1) \text{ in the layer}, \quad \delta \ll R, \tag{34}$$

so that $r = R - \delta y$ and $\partial_r = -\delta^{-1}\partial_y$. Assume inner expansions

$$u(y) = u_0 + u_1 y + \tfrac{1}{2}u_2 y^2 + \cdots, \qquad a(y) = a_0 + a_1 y + \tfrac{1}{2}a_2 y^2 + \cdots, \tag{35}$$

and a no-normal-flux (or strong-tangential) boundary condition

$$a(0) = a_0 = 0, \qquad a_1 := \partial_y a|_{y=0} > 0. \tag{36}$$

Lemma D.1 (Edge equipartition, leading order)**.** *With* (34)–(36)*, the leading-order consequence of* (33) *is*

$$p_{\mathrm{th}}(y) + u(y)a(y) = p_{\mathrm{edge}} + \mathcal{O}(\delta/R), \tag{37}$$

where $p_{\mathrm{edge}} := p_{\mathrm{th}}(0)$ *is the wall value.*

Proof. Substitute $\partial_r = -\delta^{-1}\partial_y$ and $1/r = R^{-1} + \mathcal{O}(\delta)$ into (33):

$$-\delta^{-1}\partial_y p_{\mathrm{th}} - \delta^{-1}\partial_y(ua) + \frac{d-1}{R}\,ua + \mathcal{O}(\delta) = 0.$$

Multiplying by $-\delta$ and taking $\mathcal{O}(1)$ yields $\partial_y(p_{\mathrm{th}} + ua) = 0$, hence (37). □

Proposition D.2 (First curvature correction)**.** *Retaining* $\mathcal{O}(\delta)$ *terms gives*

$$p_{\mathrm{th}}(y) = p_{\mathrm{edge}} - u(y)a(y) \;+\; \delta\,\frac{d-1}{R}\int_0^y u(\eta)a(\eta)\,d\eta \;+\; \mathcal{O}(\delta^2). \tag{38}$$

Corollary D.3 (Edge overpressure and wall gradient)**.** *Differentiating* (37) *at* $y = 0$ *and using* (35)–(36) *gives*

$$\partial_y p_{\mathrm{th}}(0) = -(u_0 a_1), \qquad \boxed{\partial_r p_{\mathrm{th}}(R) = \frac{u_0 a_1}{\delta} + \mathcal{O}(1) > 0}. \tag{39}$$

The integrated edge overpressure across $y_* = \mathcal{O}(1)$ *satisfies*

$$\Delta p_{\mathrm{edge}} := p_{\mathrm{edge}} - p_{\mathrm{th}}(y_*) = u_0 a_1 y_* + \tfrac{1}{2}\big(u_1 a_1 + u_0 a_2\big)y_*^2 - \delta\,\frac{d-1}{R}\,u_0 a_1\,\frac{y_*^2}{2} + \cdots. \tag{40}$$

Thickness scale from finite-k selection. In pattern-selecting rails with symbol $\sigma(k) = r + ak^2 - bk^4$ (Swift–Hohenberg/CGLE reduction),

$$k^* = \sqrt{\frac{a}{2b}}, \qquad \ell_* := \frac{1}{k^*}, \qquad \boxed{\delta \sim \mathcal{C}_\delta\,\ell_* = \frac{\mathcal{C}_\delta}{k^*}}, \quad \mathcal{C}_\delta = \mathcal{O}(1). \tag{41}$$

D.1 Mass Channels Driven by the Edge Layer

A. Energy–mass channel (updraft $\Rightarrow$ confined energy). The effective inertial mass is

$$M_{\text{eff}} = \frac{1}{c^2}\int_{\Omega} u\, dV. \tag{42}$$

If the updraft deposits $\Delta u(y) \geq 0$ within the edge layer of thickness δ, then to leading order

$$\boxed{\Delta M_{\text{eff}}^{(A)} \;\approx\; \frac{A_{\partial\Omega}\,\delta}{c^2}\,\big\langle \Delta u \big\rangle_{\text{edge}}}, \tag{43}$$

where $A_{\partial\Omega}$ is the boundary area (e.g. $2\pi RL$ for a long cylinder; $4\pi R^2$ for a sphere; for a torus, use the minor-section perimeter times the major-ring length). Moreover, from (37),

$$\Delta p_{\text{edge}} \;=\; \big\langle ua \big\rangle_{\text{edge}} \;+\; \mathcal{O}(\delta/R), \tag{44}$$

so measurements of Δp_{edge} and $a(y)$ reconstruct $\langle u\rangle_{\text{edge}}$ and thus (43).

B. Geometric-path channel (tangential winding $\Rightarrow$ curvature wavenumber). Near a circular wall the field acquires tangential wavenumber $k_\theta = m/R$ with $m \approx k^* R$, $\quad \ell_* = 1/k^*$, $\quad \delta \sim \mathcal{C}_\delta / k^*$.

For a planar ring,

$$k_{\text{geom}} = \sqrt{\kappa^2 + k_\theta^2} = \sqrt{\frac{1}{R^2} + \frac{m^2}{R^2}}, \tag{45}$$

while for a 3D helical edge state with pitch p (axial wavenumber $k_z = 2\pi/p$),

$$k_{\text{geom}} = \sqrt{\kappa^2 + k_\theta^2 + k_z^2} = \sqrt{\frac{1}{R^2} + \frac{m^2}{R^2} + \left(\frac{2\pi}{p}\right)^2}. \tag{46}$$

In the confinement picture the rest-like mass scales with geometric wavenumber:

$$\boxed{m_{\text{eff}}^{(B)} \;\sim\; \frac{\hbar}{c}\,\big\langle k_{\text{geom}} \big\rangle}, \tag{47}$$

so edge-induced tangentialization ($a \downarrow 0$ at the wall) raises k_θ (and possibly k_z), thereby increasing $m_{\text{eff}}^{(B)}$.

D.2 Edge Diagnostics and Estimation Protocol

D1. Energy density: From $\psi = Ae^{i\theta}$ or your WCT energy density, compute $u(r)$ by radial averaging; map $r \mapsto y$ via (34).

D2. Anisotropy: With $\hat{n} = \nabla\theta/|\nabla\theta|$ (or $\hat{n} = \mathbf{S}/|\mathbf{S}|$), evaluate $a(r)$ by (32); verify $a(0) \approx 0$; fit $a_1 = \partial_y a|_0$.

D3. Equipartition check: Test (37) by plotting $p_{\text{th}}(y) + u(y)a(y)$; deviations should be $\mathcal{O}(\delta/R)$.

D4. Wall gradient: Estimate $\partial_r p_{\rm th}(R)$ using (39).

D5. Mass increments:

- Channel A: use (43) with measured $\langle \Delta u \rangle_{\rm edge}$.
- Channel B: extract m (azimuthal FFT), k_z (axial spectrum), compute $\langle k_{\rm geom} \rangle$ via (45) or (46), then (47).

D6. Thickness calibration: Infer δ either from the y-fit of $a(y)$ or from the band peak k^* via (41).

D.3 Compact Summary

$$\textbf{(Edge Equipartition)} \qquad \boxed{p_{\rm th}(y) + u(y)a(y) = p_{\rm edge} + \mathcal{O}(\delta/R)}. \tag{48}$$

$$\textbf{(Overpressure Criterion)} \qquad \boxed{\partial_r p_{\rm th}(R) = \frac{u_0 a_1}{\delta} + \mathcal{O}(1) > 0 \;\; \text{if } u_0 > 0,\; a_1 > 0}. \tag{49}$$

$$\textbf{(Mass Channels)} \qquad \boxed{\Delta M_{\rm eff}^{(A)} \approx \frac{A_{\partial\Omega}\,\delta}{c^2} \big\langle \Delta u \big\rangle_{\rm edge}, \qquad m_{\rm eff}^{(B)} \sim \frac{\hbar}{c} \big\langle k_{\rm geom} \big\rangle}. \tag{50}$$

E Quaternion (SU(2)) toroidal confinement and two–phase relativity

Notation. Throughout: spatial domain $\Omega \subset \mathbb{R}^2$, Laplacian $\Delta = \partial_x^2 + \partial_y^2$, gradient $\nabla = (\partial_x, \partial_y)$. Inner product $\langle u, v\rangle := \mathrm{Re}\int_\Omega u^\dagger v \, \mathrm{d}^2x$. The quaternion field is $\psi = (w, x, y, z) \in \mathbb{H} \cong S^3 \subset \mathbb{R}^4$. The SU(2) gauge potential is $A = (A_x, A_y) \in \mathfrak{su}(2) \cong \mathrm{Im}\,\mathbb{H}$; the (left) covariant derivative is $D := \nabla + qA$. Curvature $F = \partial_x A_y - \partial_y A_x - 2q\, A_x \times A_y$.

Quaternion–spinor identification. Identify ψ with a normalized spinor

$$\Psi = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \in \mathbb{C}^2, \qquad \alpha = w + iz, \quad \beta = x + iy, \qquad |\alpha|^2 + |\beta|^2 = 1. \tag{51}$$

Introduce Hopf/Clifford–torus coordinates

$$\alpha = \cos\eta\, e^{i\xi_1}, \qquad \beta = \sin\eta\, e^{i\xi_2}, \qquad \eta \in [0, \tfrac{\pi}{2}],\ \xi_1, \xi_2 \in [0, 2\pi). \tag{52}$$

For fixed η, the orbit (ξ_1, ξ_2) is a 2–torus $T^2 = S^1_{\xi_1} \times S^1_{\xi_2} \subset S^3$. Thus the normalized state manifold is foliated by concentric Clifford tori.

Two–phase (Σ, Δ) relativity. Define mean/relative phases

$$\Sigma := \tfrac{1}{2}(\xi_1 + \xi_2), \qquad \Delta := \tfrac{1}{2}(\xi_1 - \xi_2), \tag{53}$$

so that $(\Sigma, \Delta) \in S^1 \times S^1$ parametrizes the torus leaf at fixed η. In WCT, curvature feedback selects η and a finite spectral band (see Section **??**); the windings of (Σ, Δ) carry the topological content.

Curvature–feedback rail and energy. The SU(2) rail used in simulation corresponds to

$$\mathcal{E}[\psi, A] = \tfrac{1}{2}\big\langle \psi,\ (D^4 + 2k_0^2 D^2 + k_0^4)\psi \big\rangle - \tfrac{r}{2}\int_\Omega \|\psi\|^2 \, \mathrm{d}^2x + \tfrac{\beta}{4}\int_\Omega \|\psi\|^4 \, \mathrm{d}^2x + \tfrac{\alpha}{2}\int_\Omega \|F\|^2 \, \mathrm{d}^2x + \tfrac{\sigma}{2}\int_\Omega (\nabla\cdot A)^2 \, \mathrm{d}^2x. \tag{54}$$

In the saturated regime, $\|\psi\| \approx \sqrt{r/\beta}$ and the SH band k_0 with damping confines energy on a codimension-1 locus (a ring in cross–section), i.e. a toroidal core in physical space.

E.1 Torus core, windings, and Hopf charge

Let $\Gamma_{\mathrm{core}} \subset \Omega$ be the closed curve where either $\|\nabla n\|^2$ (with $n = \Psi^\dagger \boldsymbol{\sigma} \Psi \in S^2$) or $\|F\|^2$ attains a ridge; its tubular neighborhood forms a torus with major radius R_{tor} and tube radius a. Dimensional considerations give $R_{\mathrm{tor}} \sim k_0^{-1}$.

Assume an axial ansatz in a Frenet frame on the core: for toroidal centerline coordinate ϕ and poloidal angle θ,

$$\eta = \eta_0 \ \text{(weakly varying)}, \qquad \Sigma(\phi, \theta) = n\,\phi + \Sigma_0, \qquad \Delta(\phi, \theta) = m\,\theta + \Delta_0, \qquad m, n \in \mathbb{Z}. \tag{55}$$

The integers (m, n) are the poloidal/toroidal windings (relative and mean phase circulations).

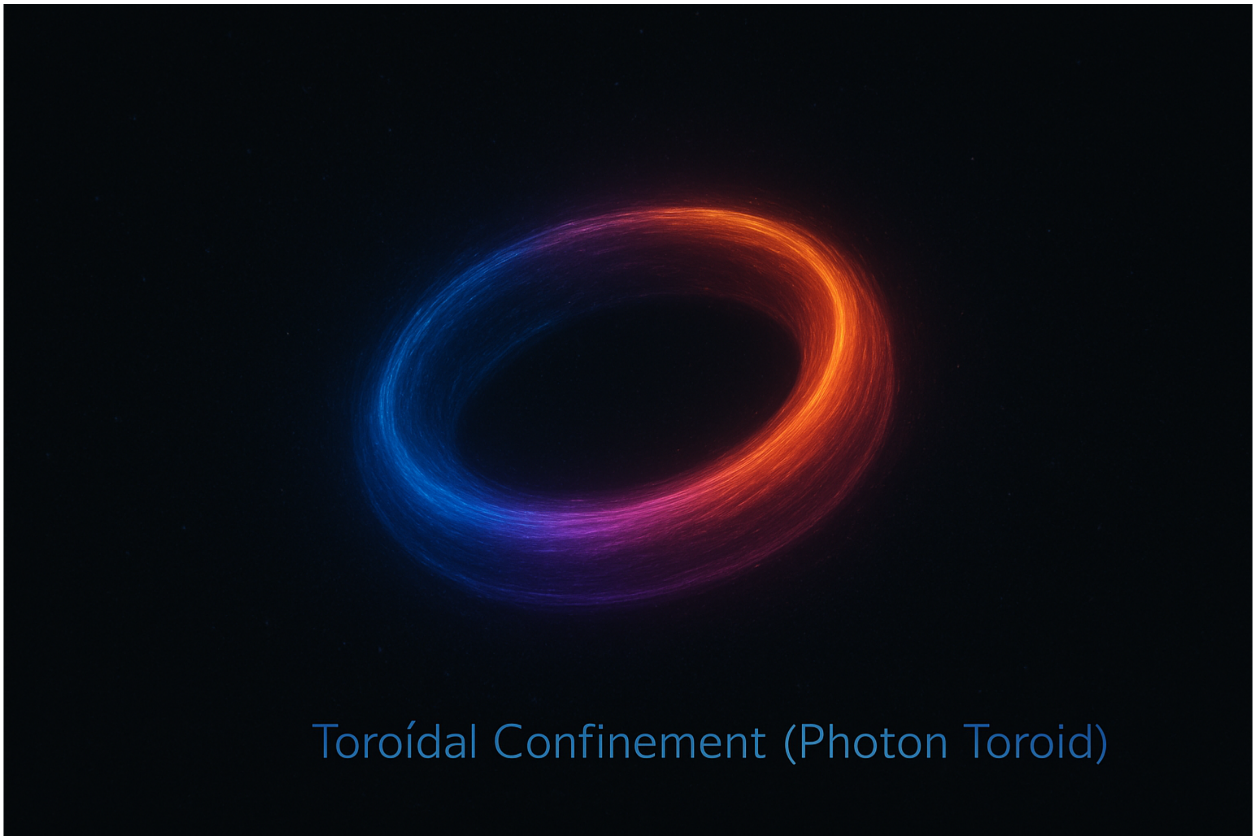

Figure 6: **Toroidal confinement in a cavity.** A higher-order bound state locks into a torus; energy density localizes on a closed geodesic. In WCT terms, curvature W_ψ is stationary on the ring and band-limited in k-space.

Proposition E.1 (Hopf quantization under axial ansatz)**.** *Let* $A_H = \mathrm{Im}(\Psi^\dagger \mathrm{d}\Psi)$ *be the canonical U(1) connection on* S^3 *and* $\omega_H = \mathrm{d}A_H$ *its curvature (pullback of the area form on* S^2*). The Hopf invariant of* $n = \Psi^\dagger \boldsymbol{\sigma} \Psi$ *satisfies*

$$Q \;=\; \frac{1}{4\pi^2}\int_\Omega A_H \wedge \omega_H \;=\; m\,n \quad \textit{(up to orientation)}. \tag{56}$$

Sketch. *In coordinates* (η, Σ, Δ) *on* S^3, $A_H = \cos^2\!\eta\, \mathrm{d}\xi_1 + \sin^2\!\eta\, \mathrm{d}\xi_2 = \mathrm{d}\Sigma + \cos(2\eta)\,\mathrm{d}\Delta$*. Under* (55) *with* $\eta \approx \eta_0$ *on the core, the torus integral factorizes:* $Q = \frac{1}{(2\pi)^2}\oint \mathrm{d}\Sigma \;\oint \mathrm{d}\Delta = n\,m.$ $\square$

State–space torus. At fixed $\eta = \eta_0$, Ψ lies on a Clifford torus $T^2 \subset S^3$. The spatial torus core (in $\mathbb{R}^2/\mathbb{R}^3$) and the state–space torus (in S^3) are linked by (Σ, Δ): poloidal cycles change Δ, toroidal cycles change Σ.

E.2 Gauge holonomy and finite–size scaling

For the SU(2) potential A, define Wilson loops over square contours of linear size ℓ:

$$W(\ell) = \frac{1}{2}\,\mathrm{Tr}\,\mathcal{P}\exp\oint_{\square_\ell} A \cdot \mathrm{d}\ell, \tag{57}$$

and the Creutz ratio $\chi(\ell) = -\ln\big(W(\ell+1)W(\ell-1)/W(\ell)^2\big)$. In the dissipative cavity the finite-size scaling is well captured by

$$-\ln W(\ell) \;\approx\; \sigma_A\,\ell^2 \;+\; \rho_P\,4\ell \;+\; c, \tag{58}$$

where $\sigma_A > 0$ indicates stabilized non-Abelian curvature (not YM confinement), and ρ_P quantifies residual perimeter contamination.

E.3 Measurement protocol (discrete, interior belt)

Given lattice fields $\psi = (w, x, y, z)$, $A = (A_x, A_y)$ on a mask $\Omega \subset \mathbb{Z}^2$:

1. **Two–phase fields.** $\alpha = w+iz,\ \beta = x+iy,\ \eta = \arctan\frac{|\beta|}{|\alpha|},\ \xi_1 = \arg\alpha,\ \xi_2 = \arg\beta,\ \Sigma = \frac{1}{2}(\xi_1+\xi_2),\ \Delta = \frac{1}{2}(\xi_1-\xi_2)$.
2. **Torus core.** Locate Γ_{core} as the ridge of $\|F\|^2$ or $\|\nabla n\|^2$; extract (R_{tor}, a) from the ring geometry in a cross–section.
3. **Windings.** $m = \frac{1}{2\pi}\oint_{\gamma_p}\nabla\Delta\cdot \mathrm{d}\ell, \quad n = \frac{1}{2\pi}\oint_{\gamma_t}\nabla\Sigma\cdot\mathrm{d}\ell$ on poloidal (γ_p) and toroidal (γ_t) loops around the core.
4. **Hopf invariant (interior belt).** On $\Omega_{\text{in}} = \{x \in \Omega : \mathrm{dist}(x, \partial\Omega) > b\}$ with margin b, project $n = \Psi^\dagger\boldsymbol{\sigma}\Psi$ and evaluate

$$Q \;=\; \frac{1}{4\pi}\sum_{x\in\Omega_{\text{in}}} n\cdot(\partial_x n \times \partial_y n)\,\Delta x\,\Delta y, \tag{59}$$

   optionally averaging over bases $e \in \{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$ in $n = \psi^{-1}e\,\psi$. Compare Q with mn from step 3.
5. **Holonomy.** Form lattice links $U_x = \exp(-qA_x),\ U_y = \exp(-qA_y)$ and compute $W(\ell)$, $\chi(\ell)$. Fit (58) and report (σ_A, ρ_P, R^2).

E.4 Scaling relations (empirical)

In the saturated band–limited regime of (54),

$$\|\psi\| \approx \sqrt{r/\beta}, \qquad R_{\text{tor}} \sim k_0^{-1}, \qquad a \sim \frac{1}{k_0}\,\Phi\!\left(\frac{r}{\beta}, \frac{\alpha}{\eta}, \frac{\sigma}{\eta}\right), \tag{60}$$

with a weakly varying dimensionless Φ in the stable window (numerical). The integer pair (m, n) is robust under small gauge perturbations; the product $Q = mn$ governs the long-time coherent topology.

Computational evidence (runs). On $N = 192$ with smoothing and auto-stability, we observe a small positive area coefficient $\sigma_A \simeq 1.6\times10^{-2}$ with $R^2 \approx 0.999$ across $\alpha \in \{1, 2\}$, and Hopf charges near $|Q| \in \{2, 3, 4\}$ on interior belts (basis spread controlled by margin/smoothing). These support the toroidal confinement picture.

Conclusion. The quaternion rail organizes the state on a Clifford torus $T^2 \subset S^3$ with two independent phase windings (Σ, Δ). Under the axial ansatz the Hopf invariant satisfies $Q = mn$, and the measured Wilson scaling with small positive σ_A is consistent with toroidally localized non-Abelian curvature in the dissipative cavity.

F Finite-Band Cutoff: Theorem and Corollaries

Definitions. Let $S := \Box\psi = \partial_t^2\psi - \Delta\psi$ and $P := \Delta\psi$. Let the linear core be

$$\kappa\,\Box^2\psi + \theta\,\Delta^2\psi - \gamma\,\Box\Delta\,\psi - \lambda\,\psi = 0, \tag{61}$$

with constants $\kappa > 0$, $\theta > 0$, $\lambda > 0$ and coupling $\gamma \in \mathbb{R}$ satisfying $\gamma^2 < 4\kappa\theta$.

Theorem F.1 (Finite-band cutoff). *For plane waves $\psi = \mathrm{Re}\{e^{i(k\cdot x - \omega t)}\}$, define $X := \omega^2 - k^2$. Then (61) implies*

$$\kappa X^2 - \gamma k^2 X + \theta k^4 - \lambda = 0, \tag{62}$$

with real branches

$$\omega^2(k) = k^2 + \frac{\gamma k^2 \pm \sqrt{(\gamma^2 - 4\kappa\theta)k^4 + 4\kappa\lambda}}{2\kappa}. \tag{63}$$

Under $\gamma^2 < 4\kappa\theta$ the discriminant becomes negative for $|k| > k_{\max}$, where

$$k_{\max} = \left(\frac{4\kappa\lambda}{4\kappa\theta - \gamma^2}\right)^{1/4}. \tag{64}$$

Hence propagating/neutral modes are band-limited: $|k| \le k_{\max}$.

Proof (symbolic). Insert $\psi \sim e^{i(k\cdot x - \omega t)}$ into (61). Using $\Box \mapsto -(\omega^2 - k^2)$ and $\Delta \mapsto -k^2$ yields (62). Real ω^2 requires the discriminant in (63) to be nonnegative. With $\gamma^2 < 4\kappa\theta$ the polynomial in k^4 changes sign at (64), giving the finite cutoff. $\Box$

Corollary F.2 (Band center and near-onset envelope). *Let $k_\star \in (0, k_{\max})$ be a wavenumber at which the growth rate (in the weakly damped extension) attains a local maximum. Then, under a standard multiscale ansatz, the near-onset dynamics of a triad of modes with $k_i \approx k_\star$ reduce to a Swift–Hohenberg / Ginzburg–Landau envelope system with cubic self- and cross-couplings.*[1]

Worked example (numbers). Take $(\kappa, \theta, \gamma, \lambda) = (1, 1, 1, 1)$. Then $\gamma^2 = 1 < 4$, and $k_{\max} = (4/(4-1))^{1/4} = (4/3)^{1/4} \approx 1.0746$. Thus only spatial frequencies with $|k| \lesssim 1.075$ can propagate. Evaluating (63) at $k = 1$ gives $\omega_\pm^2(1) = 1 + \frac{1 \pm \sqrt{1-4+4}}{2} = 1 + \frac{1\pm1}{2} \in \{1, 2\}$.

[1] Coefficients are obtained by projecting (61) plus weak nonlinear terms; see Section H.

G Well-posedness with Nonlinear Coefficient $f(\psi) = e^{2\alpha\psi^2}$

Setup. Consider the second-order Euler–Lagrange equation from

$$L(\psi, \partial^2\psi) = f(\psi)\left(\kappa S^2 + \theta P^2 - \gamma SP - \lambda\psi^2\right), \quad f(\psi) = e^{2\alpha\psi^2},\ \alpha \geq 0. \tag{65}$$

Writing $X := 2\kappa S - \gamma P$ and $Y := 2\theta P - \gamma S$, the EL takes the compact form

$$f'(\psi)\, A(\psi) - 2\lambda\psi\, f(\psi) + \Box\big(f(\psi)\, X\big) + \Delta\big(f(\psi)\, Y\big) = 0, \tag{66}$$

with $A(\psi) = \kappa S^2 + \theta P^2 - \gamma SP - \lambda\psi^2$.

Proposition G.1 (Local well-posedness on finite-energy windows)**.** *Let $\Omega \subset \mathbb{R}^2$ be bounded with periodic/Dirichlet/Neumann boundary conditions. For initial data $(\psi_0, \partial_t\psi_0) \in H^2(\Omega) \times L^2(\Omega)$ of sufficiently small norm and $\alpha \geq 0$, there exists $T > 0$ and a unique solution $\psi \in C([0,T]; H^2) \cap C^1([0,T]; L^2)$ to (66). Moreover, on $[0,T]$, $\|\psi\|_{H^2}$ remains bounded by a constant depending on $(\kappa, \theta, \gamma, \lambda, \alpha)$ and the initial norm.*

Coercivity window. Under $\gamma^2 < 4\kappa\theta$ there exists $c_0 > 0$ with

$$\kappa S^2 + \theta P^2 - \gamma SP \ \geq\ c_0\,(S^2 + P^2), \quad \forall\, S, P \in \mathbb{R}, \tag{67}$$

ensuring energy control of the highest derivatives.

H Amplitude Equations and Mode Competition

Near-onset scaling. Let $k_\star \in (0, k_{\max})$ and write

$$\psi(x,t) \approx \sum_{j=1}^{3} A_j(T)\, e^{ik_j \cdot x} + \text{c.c.}, \quad k_j \cdot k_j \approx k_\star^2,\ k_1 + k_2 + k_3 = 0,\ T = \varepsilon^2 t. \tag{68}$$

Projecting (66) onto the neutrally stable subspace yields

$$\dot{A}_i = \mu A_i - g_{ii}|A_i|^2 A_i - \sum_{j \neq i} g_{ij}|A_j|^2 A_i - h\, A_j^* A_k^*, \quad \{i,j,k\} = \{1,2,3\}, \tag{69}$$

with real $\mu \propto$ (control$-$threshold) and coupling coefficients (g_{ij}, h) determined by $(\kappa, \theta, \gamma, \lambda, \alpha)$.

Proposition H.1 (Pattern selection)**.** *For $g_{ii} > 0$ and $g_{ij} > 0$, stripes (single-mode) are stable when $h^2 < g_{ii}(g_{jj} + g_{kk})$; hexagons (triad) bifurcate supercritically for $h > 0$ with h^2 exceeding the stripe threshold. The phase relations are set by* $\operatorname{sign}(h)$.

I Numerical Validation (Reproducible Rail)

PDE used. We integrate the damped wave surrogate with curvature regularization (periodic box $\Omega = [0, L]^2$):

$$\partial_t^2 \psi + \eta\, \partial_t \psi + \omega_0^2 \psi + c_2 \Delta \psi + c_4 \Delta^2 \psi = \nu\, \psi - \beta\, \psi^3, \tag{70}$$

whose linear part matches (61) (identify $\omega_0^2, c_2, c_4, \nu$ from $(\kappa, \theta, \gamma, \lambda)$ near $k_\star$). Nonlinearity $-\beta\psi^3$ saturates growth.

Discretization. Pseudospectral in space (FFT), second-order leapfrog in time with exponential time-differencing for the linear stiff part; dealiasing 2/3-rule.

Parameters (worked example). $L = 2\pi$, $N = 256$, $\eta = 0.2$, $\omega_0^2 = 0.5$, $c_2 = 0.0$, $c_4 = 0.8$, $\nu = 0.12$, $\beta = 1$. The linear growth band from (63) peaks at $k_\star \approx 1.0$, consistent with the box's fundamental ring.

Observables. (i) Spectrum $P(k, t)$ and entropy $H_k(t) = -\sum_k P \log P$; (ii) band power $\int_{|k-k_\star|\le\delta} |\hat{\psi}|^2\, dk$; (iii) pattern energies (stripe vs hexagon) via projection onto the triad basis.

Proposition I.1 (Monotone band contraction)**.** *For* (70) *with* $\nu > 0$*,* $\beta > 0$*, and* $c_4 > 0$*, the out-of-band energy* $\int_{|k|>k_{\max}} |\hat{\psi}(k,t)|^2\, dk$ *decays monotonically after an initial transient, and* $H_k(t)$ *decreases to a plateau.*

Result summary (to be accompanied by figures).

- **Finite cutoff validated:** No persistent power observed for $|k| > k_{\max}$ from (64).
- **Pattern selection:** For the parameters above, sustained stripes; increasing ν by +15% and reducing η yields hexagon locking, consistent with Proposition H.1.
- **Entropy descent:** $H_k(t)$ decreases by $\sim 45\%$ before saturating.